

## MERN Backend - Model & Flow Notes

### 1. Backend Flow:

Frontend -> Route -> Controller -> Model -> MongoDB

### 2. Database Connection:

`mongoose.connect(...)` connects Node.js to MongoDB.

### 3. Schema:

Defines the structure and validation rules of data.

### 4. Model:

Model is created from Schema and communicates with MongoDB.

Example: `const Doctor = mongoose.model('Doctor', doctorSchema);`

### 5. Controller:

Business logic is written here.

Example: `const doctor = await Doctor.create(req.body);`

### 6. Breakdown of:

`const doctor = await Doctor.create(req.body);`

- Doctor : Mongoose Model
- `create()` : Built-in method to save a new document
- `req.body` : Data received from frontend
- `await` : Wait until data is saved
- `const doctor` : Variable that stores the saved document

### 7. CRUD Methods:

Create : `Model.create()`

Read : `Model.find()`, `Model.findById()`, `Model.findOne()`

Update : `Model.findByIdAndUpdate()`

Delete : `Model.findByIdAndDelete()`

### 8. Important Project Structure:

`config/` -> DB Connection

`models/` -> Schema & Model

`controllers/` -> Logic

`routes/` -> API Endpoints

`server.js` -> Starts server

### 9. Remember:

Model file contains Schema + Model.

`create()`, `find()`, `update()`, `delete()` are used inside Controllers.